

PATENT SPECIFICATION

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NO DRAWINGS.



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E1(P:T1B).

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COMPLETE SPECIFICATION.

Improvements in or relating to the Production of Thermoplastic Products Derived from Di-Epoxy Compounds.

We, SHELL INTERNATIONALE RESEARCH
MAATSCHAPPIJ N.V., a Company organised
under the laws of The Netherlands, of 30
Carel van Bylandtlaan, The Hague, The
5 Netherlands, do hereby declare the inven-
tion, for which we pray that a patent may
be granted to us, and the method by which
it is to be performed, to be particularly des-
cribed in and by the following statement:—

10 The invention relates to a process for the
production of a thermoplastic product suit-
able for the manufacture of threads, fibres
and similar articles. This product has a
high molecular weight and almost exclu-
sively a linear structure, and is obtained
15 by a process which comprises reacting a di-
hydroxy compound with a substantially
equimolar amount of a di-epoxy compound,
in which process either the dihydroxy com-
pound or the di-epoxy compound, or both,
20 contain an SO₂ group joining two aromatic
nuclei, in the presence of an alkaline com-
pound as catalyst, and in a solvent in which
the dihydroxy compound, the di-epoxy com-
pound and the product are at least partly
soluble and which is substantially non-re-
25 active with the dihydroxy compound or the
di-epoxy compound, the reaction mixture
being substantially free from water, the
nature and the quantity of the solvent and
the temperature being such that the reaction
30 product remains dissolved in the reaction
mixture until the product has an intrinsic
viscosity (measured in dimethyl formamide)
of at least 0.4.

35 Solvents suitable for the process according
to the present invention generally have a di-
electric constant of at least 5 at the reaction
temperature. Solvents having a di-electric
40 constant of at least 10 at the reaction tem-
perature usually give the best results.

[Price

Examples of suitable solvents are ketones,
such as methyl isobutyl ketone and cyclo-
hexanone; in particular aromatic ketones
such as acetophenone and benzophenone; 45
nitriles such as acetonitrile; phenyl aceto-
nitrile, 1,3-propane dinitrile and benzo-
nitrile; nitrocompounds such as nitroben-
zene, and 1-chloro-3-nitrobenzene; organic
sulphoxides such as dimethyl sulphoxide; 50
and sulphones such as cyclotetramethylene
sulphone (sulpholane), 3-methyl cyclotetra-
methylene sulphone and 3,4-dimethyl cyclo-
tetramethylene sulphone. Especially im-
portant sulphony compounds and nitriles 55
are mainly those in which a divalent sul-
phoxy group (SO or SO₂) or the cyano group
are directly attached to at least one carbon
atom to which one or more hydrogen atoms
are directly attached. If desired the sol- 60
vent used may be a mixture of various com-
ponents, for example a mixture of di-n-
propyl sulphoxide and diphenyl sulphoxide.
Solvents catalysing side-reactions, for ex-
ample of the epoxy groups, are undesirable. 65
Substantially equimolar quantities should
be understood as quantities in which the
molar ratio of dihydroxy compounds to di-
epoxy compound is in the range of from
about 1.04 to about 0.98, preferably from 70
about 1.02 to about 0.99. The quantity of
dihydroxy compounds present as salt, for
example alkali metal salt of a dihydric
phenol, is included in this ratio as dihy- 75
droxy compound.

There is no objection to mixing in one
operation the overall quantities of di-epoxy
compound and dihydroxy compound to be
reacted if these quantities are measured with
a sufficient degree of accuracy. Alternati- 80
vely the di-epoxy compound may at first
be reacted with a small excess of the dihy-
droxy compound and the ratio of these com-

pounds may be corrected to the equimolar ratio in a latter stage of the reaction.

The dihydroxy compound preferably used is a dihydric phenol in which the hydroxy groups, if attached to the same aromatic nucleus do not occupy the ortho-position relative to each other.

Preferred di-epoxy compounds are di-(epoxy alkyl) ethers of dihydric phenols just referred to, and particularly the diglycidyl ethers.

Very good products are obtained by reacting a diphenylol sulphone with its diglycidyl ether. Products obtained by the process according to the invention by reacting a diphenylol sulphone with the diglycidyl ethers of other dihydric phenols or else by reacting these other dihydric phenols as such with the diglycidyl ether of diphenylol sulphone are however, also very suitable. Examples of these other dihydric phenols are 2,2-di-p-phenylol-propane and also di-p-phenylol methane, hydroquinone, resorcinol, 3,3'- and 4,4'-dihydroxy diphenyl, various dihydroxy diphenyl ethanes and various dihydroxy naphthalenes. Other suitable starting materials are mixtures containing both a diphenylol sulphone and its diglycidyl ether and one or more of these other dihydric phenols and/or the glycidyl ethers thereof.

The favourable effect of the solvent used according to the invention is based on the fact that the product remains in solution until its molecular weight has increased to relatively high values. A suitable choice of solvent and its amount and also of the temperature, will prevent separation of solid macromolecular product either in a crystalline or in an amorphous state, until the product has reached an intrinsic viscosity of at least 0.4. Preferably separation is prevented until an intrinsic viscosity is reached of at least 0.8. Reasonable concentrations of the product in the reaction mixture, for example at least 5 or preferably at least 10% by weight are thus obtained. In this way even much higher intrinsic viscosities may be attained, for example 1.4. The given values of the intrinsic viscosity are determined in dimethyl formamide.

The increase of the molecular weight of the present products is accompanied by improved mechanical properties. Thus very high values of yield stress and impact strength were obtained. There is the added advantage of a high softening point and a high distortion temperature. Further these products show a high chemical resistance when contacted with acids and bases.

The product of the process according to the invention is suitable for the production of threads fibres and fabrics by drawing, spinning and weaving, and articles other than threads, fibres and fabrics by mould-

ing, extrusion injection moulding and milling.

As regards the conditions under which the reaction according to the invention is allowed to proceed, reference should also be made to the advantages of the absence of water and the purity of base materials.

The requirement for an anhydrous or substantially anhydrous medium means that the catalyst is also supplied in an anhydrous condition and that the catalyst added is a substance which does not cause water to be formed in the reaction mixture. The addition of separately prepared salts of strong bases and the dihydroxy compound being used to the reaction mixture is preferred. Examples of these strong bases are those of the alkali and alkaline earth metals and of quaternary substituted ammonia. These salts will, of course, be preferably prepared from the dihydric phenol used in the reaction. Mono-salts of the dihydric phenols give good results.

A suitable method of preparing these salts in an anhydrous state is one in which dihydric phenol is reacted with an alkali metal alcoholate and the liberated alcohol is then distilled off.

The permissible limits of the water content depend on the requirements to which the product has to conform and also on the nature of the medium. It is advisable for the water content to be kept below 2% by weight, preferably below 1% by weight, and still more preferably below 0.3% by weight, based on the initial dihydric phenol.

The salts of strong bases and dihydric phenols are generally employed in quantities in the range of from 0.01 to 10 mol.%, particularly from 0.1 to 5 mol.% based on the dihydric phenol initially present.

The commercial diphenylol sulphone mixture generally contains so many contaminants that it has to be purified before use. The contaminants are partly soluble in water, partly in aromatic solvents such as benzene and toluene in which the solubility of the diphenylol sulphone mixture itself is low.

Suitable purification methods are therefore extractions with water and aromatic solvents, preferably carried out at an elevated temperature.

The diphenylol sulphone mixture may be crystallized by dissolving it in ketones such as acetone or methyl isobutyl ketone, and adding aromatics such as benzene to the solution. This method is especially advisable when the water-soluble contaminations are absent or else after they have been removed.

A preferred method of purification is the recrystallization of the diphenylol sulphone mixture from water, especially at temperatures above 100° C. The solubility curve

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130

5% by weight H ₂ SO ₄ for 1 day	3.3%	"	"
" " " " " 7 days	5.1%	"	"
" " " " " 14 days	5.4%	"	"

EXAMPLE II

- 5 The reaction was carried out as described in Example I except that during the experiment the temperature was increased to 150° C. as soon as some crystallization occurred in order to re-dissolve the crystallizing product. Two samples then taken in the course of the process showed an intrinsic viscosity of 0.45 and 0.50 respectively. Further polymerisation was prevented in these samples by adding 2.5 grams of p,p'-diphenylol sulphone. 15
- 10 After plates were pressed from the two samples (at 230° C. respectively) the following results were obtained.

Sample No.	1	2
20 intrinsic viscosity	0.46	0.50
yield stress, kg/sq.cm.	770	775
elongation at yield stress, %	15—20	15—20
impact strength, kg.cm/sq.cm.	650	620

EXAMPLE III.

- 25 The reaction was carried out as described in Example I except that the reaction temperature was 150° C. and the reaction time 50 minutes. After this period a solution of 2.5 grams of p,p'-diphenylol sulphone in 25 grams of acetophenone was added to stop the reaction. The mixture was subsequently kept at 150° C. for a further 15 minutes and then cooled to 100° C. and diluted with dimethylformamide to a 10% solution of the polymer. Methanol was added to this solution at room temperature with stirring until the polymer had completely precipitated. The precipitate was washed with methanol and then re-dissolved in dimethylformamide this time to a concentration of 2%. Methanol was added to this new solution until light clouding occurred, whereupon the solution was poured out in distilled water with vigorous stirring. The polymer was thereby flocculated into very small particles. The solvent was removed by suction and the polymer washed with water. The resulting powder was dried in a vacuum desiccator at 100° C. The intrinsic viscosity of the product, measured in dimethylformamide, was 1.37. 70

The mixture was melted and kept in a sealed tube with the exclusion of oxygen at 105° C. for 24 hours. The intrinsic viscosity of the resultant polymer was 0.42 (measured in dimethylformamide).

EXAMPLE V.

1.00 mol of diglycidyl ether of p,p'-diphenylol propane (melting point 42° C.) was reacted with 0.97 mol. of p,p'-diphenylol sulphone in the presence of 0.05 mol. of a monosodium salt of p,p'-diphenylol sulphone in 850 grams of acetophenone in a similar manner to Example III. The product was also isolated according to Example III. The intrinsic viscosity was 0.82 measure in dimethylformamide. Tests carried out with the pressed plates yielded a tensile strength of 770 kg/sq.cm., a British Standards Izod impact strength of 33 kg.cm/sq.cm. a tensile impact strength of 300 kg.cm/sq.cm., a heat distortion temperature under a load of 18.9 kg/sq.cm. (A.S.T.M.D. 648) of 114° C. and a water absorption of 1.8% by weight after water immersion for 140 hours at 20° C. The glass transition point determined from the relationship between the modulus of elasticity and the temperature was approximately 130° C. 85 90

EXAMPLE VI.

- 55 Small plates were pressed from the crystalline powder at 290° C. This treatment reduced the intrinsic viscosity to 1.29. The yield stress of the material pressed into plates was 775 kg/sq.cm., the elongation at the yield stress 15—20%, the tensile impact strength 690 kg.cm/sq.cm. 95
- 60 1.00 mol of diglycidyl ether of p,p'-diphenylol sulphone and 1.00 mole of p,p'-diphenylol sulphone were mixed with 125 grams of sulfolane (tetramethylene sulphone) in which 0.002 mol of monosodium salt of p,p'-diphenylol sulphone were dissolved. The mixture was melted and kept in a sealed tube with the exclusion of oxygen at 105° C. for 24 hours. The intrinsic viscosity of the resultant polymer was 0.42 (measured in dimethylformamide). Tests carried out with the pressed plates yielded a tensile strength of 770 kg/sq.cm., a British Standards Izod impact strength of 33 100 105

kg.cm./sq.cm., a tensile impact strength of 300 kg.cm./sq.cm., a heat distortion temperature under a load of 18.9 kg./sq.cm. (A.S.T.M.D. 648) of 114° C. and a water absorption of 1.8% by weight after immersion for 140 hours at 20° C. The glass transition point determined from the relationship between the modulus of elasticity and the temperature was approximately 130° C.

WHAT WE CLAIM IS:—

1. A process for the production of a thermoplastic product by reacting a dihydroxy compound with a substantially equimolar amount of a di-epoxy compound, in which process either the dihydroxy compound or the di-epoxy compound, or both, contain an SO₂ group joining two aromatic nuclei, in the presence of an alkaline compound as catalyst, and in a solvent in which the dihydroxy compound, the di-epoxy compound and the product are at least partly soluble and which is substantially non-reactive with the dihydroxy compound or the di-epoxy compound, the reaction mixture being substantially free from water, the nature and the quantity of the solvent and the temperature being such that the reaction product remains dissolved in the reaction mixture until the product has an intrinsic viscosity (measured in dimethyl formamide) of at least 0.4.

2. A process as claimed in Claim 1, in which the dihydroxy compound is a dihydric phenol in which the hydroxyl groups do not occupy the ortho position relative to each other in the same aromatic nucleus.

3. A process as claimed in Claim 1 or 2 in which the di-epoxy compound is a di(epoxyalkyl) ether of a dihydric phenol in which the ether-oxygen atoms do not occupy the ortho position relative to each other in the same aromatic nucleus.

4. A process as claimed in any one of the preceding Claims in which the dihydroxy compound is p,p'-di-phenylol sulphone.

5. A process as claimed in any one of the preceding claims in which the di-epoxy compound is the diglycidyl ether of p,p'-di-phenylol sulphone.

6. A process as claimed in any one of the preceding claims in which the nature and amount of the solvent and the temperature are such that the reaction product remains dissolved in the reaction mixture until an intrinsic viscosity of at least 0.8 is reached.

7. A process as claimed in any one of the preceding claims in which the solvent used is a compound having a di-electric constant of at least 5.

8. A process as claimed in Claim 7 in which the solvent used is a compound having a di-electric constant of at least 10.

9. A process as claimed in any one of the preceding claims in which the solvent used is a ketone.

10. A process as claimed in any one of Claims 1 to 8 in which the solvent used is a nitrile.

11. A process as claimed in any one of Claims 1 to 8 in which the solvent used is a nitro compound.

12. A process as claimed in any one of Claims 1 to 8 in which the solvent used is a sulfoxide.

13. A process as claimed in any one of Claims 1 to 8 in which the solvent used is a sulphone.

14. A process as claimed in any one of the preceding claims in which the catalyst used is a salt of the dihydroxy compound.

15. A process as claimed in Claim 14 in which the catalyst used is a monoalkali salt of a dihydric phenol.

16. A process as claimed in any one of the preceding claims in which the dihydroxy compound is a diphenylol sulphone having a melting point of at least 240° C.

17. A process as claimed in any one of the preceding claims in which the dihydroxy compound is a diphenylol sulphone obtained from the commercial product by recrystallization from water.

18. A process as claimed in Claim 17 in which the diphenylol sulphone is recrystallized from water at a temperature exceeding 100° C.

19. A process as claimed in Claim 18 in which recrystallization from water has taken place in the presence of an aromatic solvent.

20. A process for the production of a thermoplastic product substantially as hereinbefore described with reference to the Examples.

21. A thermoplastic product whenever prepared according to a process as claimed in any one of the preceding claims.

22. Threads, fibres or fabrics wholly or partly manufactured from at least one thermoplastic product as claimed in Claim 21.

23. Fabrics or woven substances, wholly or partly manufactured from the threads or fibres as claimed in Claim 22.

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